

Xiang-Xiang Wang

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EDUCATION

University of Nevada, Reno

Reno, NV, USA

Ph.D. in Applied Mathematics, GPA: 4.0/4.0

Aug 2021 – May 2025 (expected)

- **Concentrations:** Computer Vision & Topological Deep Learning & Mathematical Representation of Data
- **Coursework:** Numerical Analysis and Approximation & Numerical Linear Algebra & Topology & Abstract and Real Analysis & Complex Function Theory

Shanghai University

Shanghai, China

M.Sc. in Statistics, GPA: 90.25/100

Sep 2017 – Jul 2020

- **Concentrations:** Signal Processing & Algorithm Devel & Data Analysis & Quaternion Algebra
- **Coursework:** Multivariate Statistical Analysis & Bayesian Statistics & Data Science & Advanced Mathematical Statistics & Analysis of Time Series

EXPERIENCE

Organizer

Reno, NV, USA

The 2024 Workshop Matrices and Operators

Jun 2024

- Managed all aspects including logistics, registration, tech setup, and catering.

Teaching Assistant /Mentor/PhD Student

Reno, NV, USA

University of Nevada, Reno

Aug 2021 - present

- Received zero student complaints in my classes and was rated as an excellent TA by numerous class observers.
- Mentoring three Ph.D. students with patient guidance.
- Achieved all A grades in coursework, published four papers.

Statistical Analyst

Reno, NV, USA

Provost Office of University of Nevada, Reno

Jun 2023 – Aug 2023

- Requested and collected data from various departments, ensuring comprehensive and accurate datasets.
- Prepared, processed data by SPSS/R/Python and then generated statistical reports.
- Contributed to enhancing university class design and supporting strategic academic planning.

Assistant

Chongqing, China

Hengzhen Weishi Pharmaceutical Technology Limited Company

Sep 2020 – Mar 2021

- Observed and learned data collection.
- Assisted the chairman with children's education matters.

Graduate Student Researcher

Shanghai, China

Shanghai University

Sep 2017 – Jul 2020

- Independently completed the first draft of a paper in three months.
- Developed color image processing algorithm using quaternion matrices.

PROJECTS

- **Co-Requisite Assessment Program:** Utilized established statistical analysis methods in SPSS, and Python to process and analyze data for the Co-Requisite Assessment Program. Generated detailed reports for the Provost Office, enabling data-driven decisions and program evaluation.
- **Color Image Set Recognition Project:** Designed a novel framework for color image set recognition using Quaternion Grassmannians. Developed and implemented advanced algorithms to enhance image recognition accuracy.

SELECTED PUBLICATIONS

- Eigenvalues of Quaternion Tensors with Applications to Color Video Processing – Journal of Scientific Computing (2023)
- The equivalence canonical form of five quaternion matrices with applications to imaging and Sylvester-type equations – Journal of Computational and Applied Mathematics (2021)
- Arnoldi method for large quaternion right eigenvalue problem – Journal of Scientific Computing (2020)

SKILLS

Languages: MATLAB, Python, R, SPSS, SQL, C+

Technologies: LaTeX, MS Office Suite, Github